Clinical Risk in Medical AI Induced by Data Leakage and Metric Bias

A Systematic Audit of TransUNet Models for Crohn’s Disease Segmentation in Intestinal Ultrasound

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Abstract

Artificial intelligence is increasingly presented as a promising tool for assisting the interpretation of intestinal ultrasound in Crohn’s disease. Recent studies have reported very high diagnostic performances using deep learning models, including convolutional neural networks trained to detect bowel wall thickening or inflammatory patterns in ultrasound images. For example, Carter et al. reported a CNN-based system for Crohn’s disease intestinal ultrasound achieving 90.1% accuracy, 86.4% sensitivity, 94% specificity, and an AUC of 0.9777 on a dataset of 1008 images.

However, in sequential medical imaging, high internal performance does not necessarily imply clinical reliability. Ultrasound images are often extracted from video-like acquisitions, where consecutive frames from the same patient or anatomical region can be nearly identical. If these correlated frames are randomly distributed across training, validation, and test sets, the model can exploit temporal and anatomical redundancy instead of learning disease-specific representations.

This audit investigates this risk through a controlled experimental comparison of TransUNet-based segmentation models for Crohn's disease ultrasound imaging. Two architectures were evaluated: a baseline TransUNet and a more robust TransUNet implementation based on a hybrid ResNetV2 + Vision Transformer design. Each architecture was trained and evaluated under two partitioning protocols: a leakage-prone Frame-Mix protocol and a strict Patient-Aware protocol.

Across all experiments, Frame-Mix produced artificially inflated internal Dice scores, reaching up to 93.29%, while independent holdout performance remained close to 70%. In contrast, Patient-Aware evaluation produced lower but stable internal scores that accurately reflected performance on unseen patients. These results demonstrate that the main clinical risk is not simply poor model performance, but misleading evaluation: a model may appear highly reliable while silently missing a substantial portion of pathological tissue.

This work argues that strict patient-level separation and independent holdout testing are mandatory for clinically meaningful evaluation of medical AI systems in sequential ultrasound imaging.

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1. Introduction

Intestinal ultrasound has become an increasingly important imaging modality for the diagnosis and monitoring of inflammatory bowel disease, especially Crohn's disease. Its clinical value comes from its ability to visualize bowel wall structure, inflammatory thickening, and disease-related complications without radiation exposure or invasive procedures.

One of the most important ultrasonographic markers of Crohn's disease activity is bowel wall thickening. In clinical practice, bowel wall thickness above approximately 3 mm is commonly used as a surrogate marker of transmural inflammation, while normalization of bowel wall thickness is associated with transmural healing and treatment response.

Despite its clinical utility, intestinal ultrasound remains highly operator-dependent. Image acquisition and interpretation require experience, and the growing use of intestinal ultrasound creates a need for tools capable of supporting less experienced operators. This explains the increasing interest in artificial

intelligence systems for automatic detection, classification, and segmentation of Crohn's disease features in ultrasound images.

Recent work illustrates this trend. Carter et al. developed a machine-learning system based on a pretrained convolutional neural network to distinguish bowel wall thickening from normal intestinal ultrasound images. Their dataset contained 1008 images, split into 805 training images and 203 test images, and the reported results showed high classification performance, including 90.1% accuracy and an AUC of 0.9777.

Such results are clinically promising, but they also raise an important methodological question: can high internal metrics be trusted when the dataset originates from sequential ultrasound acquisitions?

This question is central because ultrasound images are not independent samples in the same way as natural images. They are often extracted from continuous acquisition sequences, where adjacent frames share the same patient anatomy, probe position, acoustic artifacts, gain settings, speckle texture, and pathological region. As a result, two frames from the same sequence may be visually almost identical.

If a dataset is split randomly at the image level, frames from the same patient or sequence may appear simultaneously in the training and test sets. This creates a form of data leakage: the model is evaluated on images that are not clinically independent from those seen during training. In such a setting, high performance may reflect memorization of patient-specific visual signatures rather than true recognition of Crohn's disease.

The goal of this audit is therefore to quantify the clinical danger of misleading metrics in medical AI. The study does not aim to prove that deep learning is ineffective. Instead, it shows that evaluation methodology can transform an imperfect model into an apparently excellent one, simply by allowing leakage between training and test data.

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2. Problem Statement and Hypotheses

The central issue addressed in this work is the **reliability of evaluation protocols in medical image segmentation**, particularly in the context of sequential ultrasound imaging.

While many recent studies report high performance metrics, these results are often obtained under evaluation conditions that do not guarantee statistical independence between training and test data. In ultrasound imaging, this problem is amplified by the intrinsic nature of the data.

2.1 The Performance Paradox

In the current literature, segmentation and classification models applied to medical imaging frequently report:

- Dice scores above **85%–90%**
- High sensitivity and specificity
- Near-perfect ROC-AUC values

However, when similar models are evaluated on **new, unseen patients**, performance typically drops significantly.

This discrepancy leads to a critical paradox:

> A model can appear clinically reliable during evaluation while failing in real-world deployment.

2.2 Ultrasound Imaging vs Natural Images

This paradox originates from a fundamental mismatch between:

- **Standard machine learning assumptions**
- **The physical reality of ultrasound imaging**

Natural Image Datasets (e.g., ImageNet)

- Images are independent samples
- Random splitting is statistically valid
- No temporal or spatial correlation between samples

Ultrasound Imaging

- Images are extracted from **continuous acquisition sequences**

- Frame rate: typically **15–30 frames per second**
- Consecutive frames are **almost identical**

As a result:

- Multiple images in the dataset represent the **same anatomical structure**
- Redundancy is extremely high
- Data independence is violated by default

2.3 Data Leakage Mechanism

When a dataset is split at the **frame level** (Frame-Mix protocol), the following occurs:

- Frames from the same patient appear in:
  - Training set
  - Validation set
  - Test set
- The test set is no longer independent

This leads to **data leakage**, defined as:

> The presence of information in the test set that was already accessible during training.

2.4 Consequences on Model Behavior

Under data leakage, the model does not learn clinically meaningful features. Instead, it exploits shortcuts:

- Ultrasound texture patterns (speckle noise)
- Probe orientation and acquisition angle
- Patient-specific anatomical structures
- Temporal continuity between frames

This results in:

- Artificially high internal performance
- Poor generalization to unseen patients

2.5 Clinical Interpretation Failure

From a clinical perspective, this behavior is dangerous.

The model:

- Produces **confident predictions**
- Appears accurate on evaluation data
- Fails to capture the full extent of pathology in new patients

This creates a **false sense of reliability**.

2.6 Hypotheses

This audit is based on the following hypotheses:

Frame-Mix Bias

A model trained with a Frame-Mix protocol will:

- Achieve high internal Dice scores
- Fail to generalize to independent patients

Patient-Aware Reliability

A model trained with a Patient-Aware protocol will:

- Show lower internal performance
- Maintain stable performance on unseen patients

Capacity Amplification Effect

Increasing model complexity (e.g., ResNetV2 + ViT):

- Will not improve generalization under strict evaluation
- Will amplify overfitting under data leakage

2.7 Objective of the Audit

The objective of this study is to:

- Quantify the gap between **reported performance** and **real clinical performance**
- Identify the impact of **data partitioning strategies**
- Evaluate the reliability of commonly reported metrics

2.8 Key Research Question

> Can we trust high-performance metrics in medical AI if the evaluation protocol does not enforce strict patient-level independence?

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3. Materials and Methods

This section describes the dataset, partitioning strategies, model architectures, and training pipeline used to evaluate the impact of data leakage on segmentation performance.

3.1 Dataset Description

The dataset consists of manually segmented intestinal ultrasound images acquired from patients with suspected or confirmed Crohn's disease.

- **Total images:** 17,488
- **Pathological (TP):** 11,396
- **Non-pathological (TNP):** 6,092
- **Total patient/sequence groups:** 50

Each group corresponds to a coherent acquisition sequence from a single patient.

3.2 Independent Holdout Set

To ensure an unbiased clinical evaluation, a subset of patients was isolated before any data splitting.

- **Patients:** 4 groups
- **Images:** 4,232
- **Pathological:** 2,882
- **Non-pathological:** 1,350

This dataset is used exclusively for **final evaluation**.

> No image from these patients is used during training or validation.

3.3 Partitioning Protocols

Two distinct data partitioning strategies were evaluated.

3.3.1 Patient-Aware Protocol (Clinically Valid)

- Data split at the **patient/sequence level**
- Each patient appears in only one set

Distribution:

- Train: 34 groups
- Validation: 4 groups
- Test: 8 groups

Properties:

- No overlap between sets
- True simulation of clinical deployment

3.3.2 Frame-Mix Protocol (Leakage-Prone)

- Data split at the **image level**
- Random shuffle followed by 80/10/10 split

Distribution:

- Train: images from 46 groups

- Validation: images from 45 groups
- Test: images from 45 groups

Properties:

- Strong overlap between sets
- Massive data leakage

3.4 Model Architectures

Two variants of TransUNet were evaluated.

3.4.1 Baseline TransUNet

- Custom CNN encoder
- Transformer:
  - 8 layers
  - 4 attention heads
  - Embedding dimension: 1024
- Standard U-Net decoder

3.4.2 Robust TransUNet (Paper Implementation)

- ResNetV2-50 backbone
- Vision Transformer:
  - 12 layers
  - 12 heads
  - Embedding dimension: 768
- DecoderCup architecture

3.5 Training Configuration

All models were trained under identical conditions.

- **Input size:** 224×224
- **Normalization:** [0, 1]
- **Augmentation:**
  - Random rotations
  - Horizontal and vertical flips
- **Optimizer:** SGD
- Momentum: 0.9
- Weight decay: 1e-4
- Learning rate: 0.01
- **Scheduler:** Polynomial decay

Loss function:

```text
Loss = 0.5 × CrossEntropy + 0.5 × DiceLoss
```

3.6 Evaluation Pipeline

Evaluation is performed in three stages:

1. **Validation monitoring (during training)**
2. **Threshold-based test evaluation**
3. **Independent holdout evaluation**

3.7 Metrics

The following metrics were used:

- Dice coefficient
- Intersection over Union (IoU)
- Precision
- Recall

Accuracy was intentionally excluded due to class imbalance.

4. Results

The results demonstrate a clear divergence between internal evaluation and real-world performance depending on the data partitioning strategy.

4.1 Internal Performance (Frame-Mix)

Under Frame-Mix:

- Dice scores reach up to **93.29%**
- Precision remains high (>90%)
- Recall appears strong on internal test sets

However, these results are misleading.

4.2 Independent Holdout Performance

On unseen patients:

- Dice drops to approximately **70%**
- Recall collapses to **~57–62%**
- Precision remains high (>90%)

4.3 Performance Gap

The difference between internal and external evaluation reaches:

- **Up to -23 Dice points**

This gap directly quantifies data leakage.

4.4 Patient-Aware Stability

Under Patient-Aware protocol:

- Internal Dice: ~67–70%
- Holdout Dice: ~68–70%

No significant gap is observed.

4.5 Key Observation

> Patient-Aware evaluation predicts real clinical performance, while Frame-Mix evaluation creates an illusion of performance.

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5. Analysis and Interpretation

The experimental results reveal a fundamental distinction between **apparent performance** and **true clinical performance**, driven entirely by the data partitioning strategy.

5.1 Data Leakage Signature

The most direct indicator of data leakage is the **performance gap between internal test and independent holdout**.

- Frame-Mix:
  - Internal Dice: up to **93.29%**
  - Holdout Dice: ~**70%**
  - Gap: **−23.12%**
- Patient-Aware:
  - Internal Dice: ~**67–70%**
  - Holdout Dice: ~**68–70%**
  - Gap: **≈ 0%**

Interpretation

- A large negative gap indicates **evaluation bias**
- A near-zero gap indicates **reliable generalization**

> The gap itself becomes a quantitative measure of data leakage.

5.2 Memorization vs Generalization

Under Frame-Mix, the model is exposed to highly correlated samples across all splits.

This leads to:

- Learning of **low-level visual signatures**
- Memorization of:
  - Speckle patterns
  - Probe orientation
  - Patient-specific anatomy

Instead of:

- Learning disease-related features
- Detecting inflammation patterns

Consequence

> The model behaves as a **lookup system**, not a diagnostic system.

5.3 Epoch Paradox

A critical observation is the divergence between:

- Internal test performance
- Independent holdout performance

Across training epochs.

Observed Behavior (Robust Model)

- Epoch 30:
  - Internal Dice: ~92.45%
  - Holdout Dice: ~73.66%
- Epoch 50:

- Internal Dice: ~93.29%
- Holdout Dice: ~70.17%

Interpretation

- Internal performance **improves**
- Real performance **degrades**

This indicates:

- Transition from partial generalization → full memorization
- Optimization driven by dataset-specific patterns

> The model becomes better at the dataset, worse at the task.

5.4 Threshold Sensitivity

Under Patient-Aware protocol:

- Optimal threshold: **0.1**
- Performance decreases significantly at standard thresholds (0.5)

Interpretation

- The model is **uncertain**
- Predictions are **low confidence**

Under Frame-Mix:

- Performance remains high across thresholds
- Predictions are artificially confident

> Confidence becomes a byproduct of leakage, not correctness.

5.5 Precision–Recall Imbalance

Across all configurations on independent data:

- Precision: **>90%**

- Recall: **~57–62%**

Interpretation

- The model is **conservative**
- It avoids false positives
- But misses large portions of the lesion

5.6 Clinical Meaning

From a clinical perspective:

- High precision → predictions are trustworthy when present
- Low recall → incomplete lesion detection

Result

> The model segments the **core** of inflammation but ignores **diffuse margins**

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6. Clinical Risk Analysis

The consequences of misleading evaluation are not theoretical. They directly translate into clinical risk.

6.1 False Confidence Scenario

A model validated under Frame-Mix reports:

> **Dice = 93.29%**

A clinician interprets this as:

- Near-perfect segmentation
- High reliability
- Safe clinical use

6.2 Real-World Behavior

On unseen patients:

- Dice: ~70%
- Recall: ~58%

This implies:

- ~40% of lesion area is not detected
- Segmentation is incomplete

6.3 Silent Failure Mode

The most dangerous aspect is that:

- The model does not produce obvious errors
- Predictions appear clean and precise

Effect

- The clinician is **not alerted**
- The error is **systematic but invisible**

6.4 Clinical Consequences

This leads to:

- Underestimation of disease extent
- Misinterpretation of severity
- Potential treatment errors

6.5 Key Insight

> A misleadingly high metric is more dangerous than a low metric.

Because:

- Low performance → lack of trust
- Inflated performance → false trust

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7. Discussion

7.1 Relation to Existing Literature

Recent studies in medical AI, including those focusing on Crohn's disease ultrasound, report high diagnostic performance using deep learning models.

For example:

- CNN-based systems achieving >90% accuracy
- High ROC-AUC values (>0.95)

These results are promising but must be interpreted in context.

7.2 Hidden Assumption

Many studies assume:

- Data independence
- Validity of random splitting

However, in ultrasound:

- This assumption is **invalid**

7.3 Implicit Risk

If frame-level splitting is used:

- Reported performance may reflect:
  - Temporal correlation
  - Patient-specific redundancy

Rather than:

- True disease detection capability

7.4 Contribution of This Work

This study does not introduce a new model.

Instead, it provides:

- A **methodological audit**
- A **quantitative demonstration** of data leakage effects
- A **clinical interpretation** of metric bias

7.5 Key Message

> Evaluation methodology defines the meaning of the metric.

Without strict protocol:

- Metrics lose clinical relevance

7.6 Limitations

All models in this study exhibit:

- Overfitting
- Limited generalization capacity

Due to:

- Dataset size
- Limited augmentation
- Training from scratch

7.7 Important Clarification

These limitations:

- Affect all configurations equally
- Do not invalidate the conclusions

Because:

- The study compares **relative behavior under different protocols**

7.8 Future Work

Improvements may include:

- Larger datasets
- Stronger data augmentation
- Domain-specific pretraining

However:

> These improvements do not eliminate the need for proper evaluation protocols.

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8. Conclusion

This study demonstrates that:

- High performance metrics in medical AI can be misleading
- Data leakage is a primary source of metric inflation
- Model capacity amplifies, rather than solves, this issue

Final Statement

> A model evaluated without strict patient-level separation cannot be considered clinically reliable.

Core Insight

- The limitation is not the model
- The limitation is the **evaluation methodology**

Closing Remark

> When evaluation is flawed, the model does not become better—it becomes dangerously convincing.

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9. Extended Results: Quantitative Evidence Across the Pipeline

To fully characterize the behavior of the model, performance was analyzed at multiple stages of the evaluation pipeline.

9.1 Training Dynamics

The training process shows a smooth and continuous improvement:

- Train Dice: **63% → 95%**
- Validation Dice: **77% → 93%**

Observation

- Training and validation curves remain tightly coupled
- No visible divergence typically associated with overfitting

Interpretation

> The absence of a train/validation gap is misleading and results from **data leakage in the validation set**.

9.2 Internal Test Evaluation (Threshold Optimization)

Performance on the internal test set (Frame-Mix) confirms extremely high scores:

Epoch	Best Threshold	Dice	Precision	Recall
30	0.5	92.45%	92.11%	93.45%
40	0.5	93.07%	92.95%	93.76%
50	0.4	93.29%	92.77%	94.38%

Observation

- Performance is:
  - High
  - Stable across thresholds
  - Consistent across epochs

Interpretation

> The model has effectively **seen the test distribution during training**.

9.3 Independent Holdout Evaluation

The independent dataset reveals the true behavior:

Epoch	Dice	Precision	Recall
20	68.87%	88.61%	57.69%
30	73.66%	89.56%	62.63%
40	72.24%	90.09%	61.22%
50	70.17%	91.03%	57.88%

Key Observations

1. **Performance Peak at Epoch 30**
   - Best generalization occurs before full convergence
2. **Degradation After Epoch 30**
   - Continued training harms generalization
3. **Precision Stability**
   - Remains consistently high (>88%)
4. **Recall Collapse**
   - Remains limited (~57–62%)

Interpretation

> The model transitions from partial generalization to **over-specialization on training data**.

9.4 Pixel-Level Imbalance Analysis

From independent evaluation:

- Predicted positive pixels: **~17–19M**
- Ground truth positive pixels: **~28M**

Interpretation

- The model under-segments lesions
- Significant portion of pathology is not detected

9.5 Patient-Level Consistency

Across patients:

- Dice remains stable:
  - ~68–74% depending on subgroup

Interpretation

- Failure is **systematic**, not case-specific
- The model consistently underestimates lesion extent

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10. Failure Mode Characterization

10.1 Nature of the Error

The model does not:

- Produce noisy predictions
- Generate random false positives

Instead, it:

- Produces **clean but incomplete segmentations**

10.2 Spatial Behavior

Observed segmentation pattern:

- Accurate detection of:

    - High-contrast lesion core
- Failure to detect:
    - Low-contrast boundaries
    - Diffuse inflammation

10.3 Root Cause

This behavior is consistent with:

- Training on **highly redundant frames**
- Lack of exposure to:
    - Anatomical variability
    - Inter-patient differences

10.4 Model Strategy

The model implicitly learns:

- Conservative decision boundaries
- High-confidence predictions on familiar patterns

10.5 Consequence

> The model prioritizes precision at the expense of recall.

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11. Methodological Implications

11.1 Evaluation Pipeline Design

A robust evaluation must include:

1. Training/validation monitoring
2. Internal test evaluation
3. Independent external validation

11.2 Single-Metric Evaluation is Insufficient

High Dice alone cannot guarantee:

- Clinical validity
- Robust generalization

11.3 Importance of Independent Data

Only the independent holdout:

- Breaks correlation
- Reveals true performance

11.4 Recommended Practice

- Always enforce:
  - Patient-level separation
- Always report:
  - Internal vs external gap

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12. Practical Recommendations

12.1 For Researchers

- Avoid frame-level random splits
- Validate on independent patients
- Report recall alongside Dice

12.2 For Reviewers

- Question unusually high performance (>90%)
- Verify dataset construction methodology

12.3 For Clinicians

- Treat reported metrics with caution
- Prefer models validated on:
  - External cohorts
  - Diverse populations

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13. Final Takeaways

Key Observations

- Data leakage can inflate Dice by **+20 points**
- Model capacity amplifies this effect
- True performance remains around **~70% Dice**

Core Message

> The reliability of a medical AI system is determined more by its evaluation protocol than by its architecture.

Final Warning

> A model that appears highly accurate under flawed evaluation can become a source of clinical error.

Code and Data Availability

The full implementation of the training pipeline, evaluation scripts, and experimental results is publicly available:

```
https://github.com/NicoDumetz/Medical-Segmentation-IA-Leakage
-Audit
```

The repository includes:

- Full training scripts (`train.py`)
- Evaluation pipeline (`threshold_test.py`, `independent.py`)
- Dataset construction scripts
- Raw logs and reproducible results

All experiments can be reproduced using the configurations provided.